Astra failure runs

Oct 15, 2026 · @Ayush

Summary

GPT-6 Astra fails reliably on one task type: analysing large rigid-jointed frames (36 to 55 members) by reasoning alone. It declined all 10 such tasks, returning no displacements or stresses.

Every other recorded miss either did not reproduce or traced back to our own task, reference, harness or a restricted protocol. With tools and feedback, Astra passed every complex CAD+FEM task tried: contact, large deflection, plasticity, buckling, transient heat, fracture, an industrial bracket and a mechanism design.

A run counts as a failure only when Astra's own answer is wrong, unusable or withheld, the reference is independently verified, and the task is solvable within the run's limits.

Confirmed failures

Fifteen failures are admitted; only the frame refusals reproduce. The CAD-edit and pipe-network cases are single observations that other Astra attempts got right.

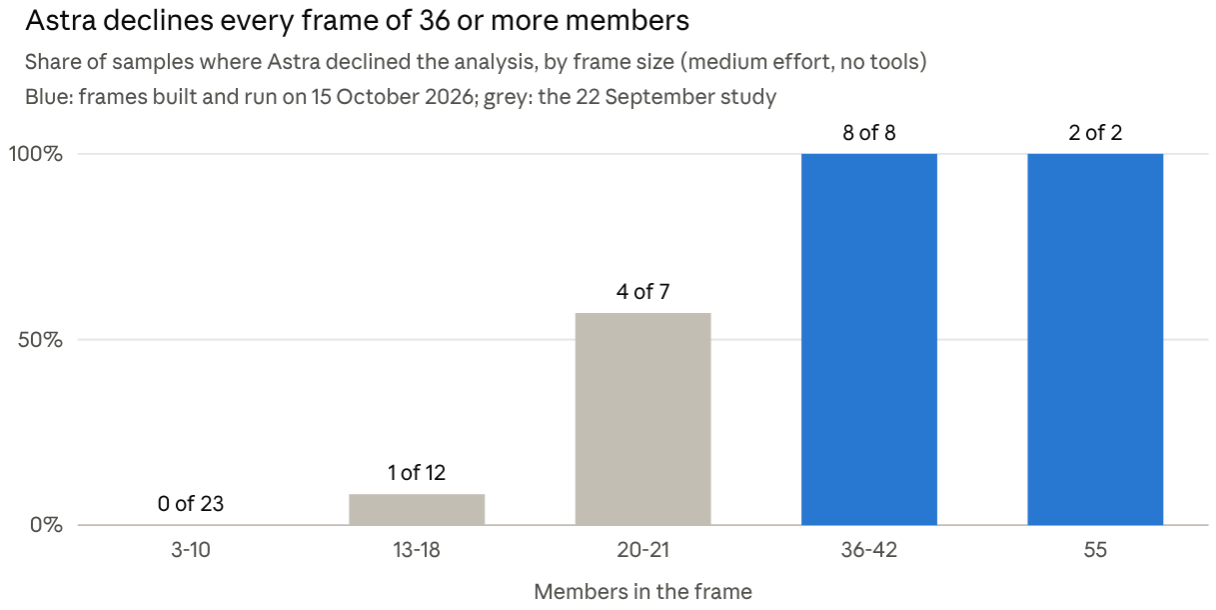
Case	Domain	Protocol	What Astra did	Reproduced
10 large frames, 36-55 members	Frame drawing + member stresses	Single call, no tools	Drew the frame, then declined the analysis (status "unavailable", no numbers)	10 of 10
Frame threebay_three_edit, 21 members	Frame drawing + member stresses	Single call, no tools	Declined the analysis	3 of 4
CAD edit d000c8c0, propeller boss	One-shot CadQuery patch	Single call, no tools	Patch does not run; the repaired answer leaves the boss floating 1.144 mm above the hub	Repeated in 3 of 4 runnable drafts

Case	Domain	Protocol	What Astra did	Reproduced
CAD edit 511b48e9, connecting-rod chamfer	One-shot CadQuery patch	Single call, no tools	Chamfer rejected by the CAD kernel, no solid	No: another draft and one feedback round succeed
CAD edit 5eb603a4, twisted bracket	One-shot CadQuery patch	Single call, no tools	JSON patch cut off before its closing brackets	No: six other drafts run
Pipe network 0003371e edit-4	Drawing edit + hydraulic analysis	Single call, no tools	Reported "pass" at J4 = 20.04 m; true 19.23 m, below the 20 m minimum	No: correct on a second sample

One further frame case (twobay_four_generate, 20 members) is unconfirmed: its confirmation samples were lost to API errors.

Large-frame refusals

Astra declined all 10 frames of 36 to 55 members, against 1 of 35 samples below 20 members.



runs/eng-frame-aggregate.json (22 Sep 2026) and runs/astra-frames*-20261015 (15 Oct 2026); one sample per task unless noted

Each task gives a dimensioned frame and asks for an SVG drawing plus the probe displacements and the peak member-end stress. References come from a linear frame solver, checked against a 4x member subdivision and global equilibrium. In every refusal Astra drew the frame correctly and stated that the stiffness equations had not been solved; it never reported wrong numbers. With tools the same analyses are easy, so this failure is specific to unaided analysis of large coupled systems: a 78-member pin-jointed truss and an 8-storey modal problem were solved exactly.

Task	Members	Indeterminacy	Cost (USD)
fourbay_four_generate	36	48	0.50
fourbay_four_edit	36	48	0.57
sixbay_three_generate	39	54	0.58
sixbay_three_edit	39	54	0.62
twobay_eight_generate	40	48	0.59
twobay_eight_edit	40	48	0.62
threebay_six_generate	42	54	0.71
threebay_six_edit	42	54	0.67
fivebay_five_generate	55	75	0.60
fivebay_five_edit	55	75	0.80

Apparent failures that were our defects

Most recorded Astra misses were our errors: in several, Astra's answer was the physically correct one.

Case	Looked like	Actual cause	How it was confirmed
CADFEM A-CADWORLD-001	Displacement 35 % low	Prompt said first-order elements; the publisher model is second-order	Independent solve: first-order 20.27 mm (Astra's answer), second-order 31.03 mm (published 31.1)

Case	Looked like	Actual cause	How it was confirmed
32 of 35 audited CAD-edit misses	Wrong geometry	14 reference defects, 2 impossible tasks, 16 ambiguous instructions	Astra matched the reference program exactly (IoU 1.000000) on 7; six references contain floating parts
SimJEB brackets 101, 108, 62	Displacements about 2x the published values	Prompt gave bolt 1 as 10.3124 mm; the STEP hole is 10.668 mm, so Astra left bolt 1 unconstrained. Published OptiStruct values are first-order and 5-10 % stiff	0 of 517 bolt-1 nodes fixed; with the prompt corrected Astra is within 1 % of the converged solution
Plate girder buckling (single-shot attempt)	Lambda 1.92 vs 0.412	Restricted protocol (one shot, low reasoning, no repair); CalculiX skips sub-unit buckling factors	Agentic run: 0.414, +0.5 %
9 single-shot script failures (inverted meshes, parser, type error)	No results	One shot with no execution feedback	6 recovered with feedback; the rest were SimJEB (above) or the girder
Scissor-jack replay crash	Clean re-run failed	Harness ran python -I, which blocks helper modules the contract allows	Fixed to -E -s -B; the saved code then passes all 7 checks
Frame drawings marked unscored_format	Missing mm units	Scorer wanted the unit on every dimension label, not once in the notes	Adjudicated as standard drafting practice

Where Astra passed

With tools Astra passed every complex task tried, often by writing the solver itself. Without tools it still solved moderate problems exactly.

Task	Physics	Protocol	Result	Cost (USD)
CADFEM-Bench v1.5, 7 valid tasks	Static, buckling, plastic limit, transient heat, fracture	Agentic	7 of 7 within tolerance	9.98 incl. discarded attempts
Plate girder EC3	Elastic buckling, thin plates	Agentic	Lambda 0.414, +0.5 %	1.14
Scissor jack, 1.5 t, tight mass budget	Mechanism design: mates, motion, strength, mass	Agentic	All 7 grader checks; 2.414 kg against a 2.62 kg product	1.24
SimJEB bracket 101, corrected prompt	Four load cases on an industrial bracket	Agentic	Within 1 % of the converged solution	1.78
Hertz contact, hemisphere on a rigid flat	Nonlinear contact	Agentic	Radius +0.07 %, peak pressure -0.09 %, own active-set solver	0.41
Large-deflection cantilever	Geometric nonlinearity (82 degree tip rotation)	Agentic	+0.02 % against the exact elastica	0.28
15 ghost-geometry CAD traps	Edits that depend on prior cuts, tapers, chamfers, shells	One-shot	13 pass; 2 are judgement calls	0.24
78-member 3D truss, 8-storey modes, portal buckling, L-fin heat	Statics, dynamics, stability, conduction	Single call, no tools	4 of 4 correct (errors 0.0-0.1 %)	1.62
3 near-threshold pipe networks	Hydraulic analysis	Single call, no tools	3 of 3 correct	1.59

Method and spend

All runs used gpt-6-astra at medium reasoning effort under a hard USD cap, $24.53 in total across seven rounds.

- Agentic protocol: tools (bash, Python with CadQuery, gmsh, scikit-fem) in a sandbox with no network; the submitted main.py is re-run from scratch (600 s, 8 GB) and only that run is graded.
- Single-call protocol: one request, no tools, 6,000 to 16,000 output tokens; the answer is parsed and graded.
- Every reference was checked before any spend: published values, closed forms, or our own solves with equilibrium, refinement or mesh-convergence checks.
- Costs use $12.50 per million uncached input tokens, $1 cached and $50 output.

Round	Calls	Cost (USD)	Outcome
CADFEM-Bench v1.5 hosted run	13 tasks attempted	9.98	7 of 7 valid pass; 1 task defect
Pipe-network confirmation	3	1.59	3 of 3 correct; the earlier miss did not reproduce
Plate girder + scissor jack	2	2.38	Both pass
SimJEB bracket, corrected	1	1.78	Pass; two task defects found
Ghost-geometry traps + 2 large frames	17	1.33	Traps pass; both frames declined
8 large frames + Hertz + elastica	10	5.85	Frames declined 8 of 8; both nonlinear tasks pass
Varied no-tools tasks	4	1.62	4 of 4 correct

Codex ran its own single-shot failure hunt and rechecks separately, about $8.10, ending with zero confirmed capability failures.

Evidence and files

All paths are under ~/SVG/cvpr2027; raw requests, responses and replays are kept for every call.

- Failure-case package: output/datasets/astra-failure-cases-v0.1 (15 admitted, 36 excluded, with reasons)

- Spend ledger: runs/hosted-v1.5/SPEND.md
- Large frames: data/eng-frame-bench-xxl, data/eng-frame-bench-xxl2, runs/astra-frames10-20261015
- Nonlinear and SimJEB runs: runs/astra-complex-20261015 (validation scripts in validation/)
- Varied no-tools tasks: runs/astra-variety-20261015
- Ghost-geometry traps: runs/astra-ghost-geometry-20261015
- Scissor-jack benchmark: output/datasets/mechdesign-bench-v0.1
- Large-deflection reference: SOFiSTiK verification BE7 (PDF)